

IK 522: Ancient Indian Chemistry and Metallurgy

Course Code: IK 522

Course Name: Ancient Indian Chemistry and Metallurgy

L-T-P-C: 3-0-0-3

Prerequisites: None

Intended for: BTech/MTech/MS/MSc/MA/Ph.D.

Mutual Exclusion: None

Approval: 64th BoA

Preamble

This course explores the development of chemical knowledge and metallurgical practices in ancient India through a structured integration of philosophical concepts, empirical observations, and technological applications. It highlights how early Indian thinkers conceptualized matter, transformation, and energy, and how these ideas translated into practical domains. By combining theoretical frameworks with technological advancements such as metal extraction, alloying, and chemical formulations, this course provides a holistic view of chemistry and metallurgy as practiced in ancient India, while also drawing connections to contemporary scientific approaches.

Course Modules with Quantitative Lecture Hours

- **Unit 1: Concepts of Matter, Energy, and Transformation:** Sāṅkhya–Pātañjala framework: Prakṛti as the fundamental constituent and its transformations. Conservation and transformation of energy, causation, and principles of collocation, storage, and dissipation of energy. Evolution of matter and infra-atomic concepts. Vedantic concept of Pañcīkaraṇa and structured formation of matter. Atomic theories in Buddhist and Jain traditions. Discussion of elements, compounds, and chemical analysis and synthesis, with connections to modern atomic theory, thermodynamics, and phase transformations. (10 hours)
- **Unit 2: Chemical Concepts in Medical and Material Traditions:** Chemistry in ancient Indian medical schools. Physical characteristics of Bhūtas and Mahābhūtas. Mechanical mixtures and chemical combinations. Emergence of molecular properties in compounds. Chemistry of colors, pigments, and material transformations. Systems of measurement including weight and capacity. Concept of minimum visible size and observational limits. Connections to formulation chemistry, materials characterization, and pigment science. (7 hours)
- **Unit 3: Nyāya–Vaiśeṣika Theory and Chemical Interactions:** Theory of atomic combinations and classification of compounds (mono- and hetero-bhautika). Concept of dynamic contact (Viśṭambha) and chemical bonding analogies. Role of heat in chemical transformations. Three-axis representation of compounds (Vācaspati model). Concept of molecular motion (Parispanda) and its relevance to kinetics. Interpretation in light of modern chemical bonding, reaction mechanisms, and molecular motion. (7 hours)
- **Unit 4: Applied Chemistry in Classical Texts and Practices:** Chemical knowledge in texts such as Bṛhatsamhitā. Preparation of Vajralepa and Vajrasamghāta as examples of coatings and composite materials. Gandhayukti as formulation science involving fragrances and mixtures. Discussion of combustion-related formulations in alchemical texts, including saltpetre (śora) based mixtures and compositions analogous to gunpowder. Broader discussion of traditional chemical processing techniques and their relevance to coatings, organic chemistry, and materials engineering. (5 hours)

- **Unit 5: Metallurgical Traditions and Technologies:** Metallurgical knowledge from texts such as Arthaśāstra and Rasashastra literature. Extraction and processing of metals including gold, silver, copper, iron, tin, mercury, lead, and zinc. Alloying, high-temperature processes, and furnace technologies. Central role of mercury (Rasa) in Indian alchemical traditions, including its purification (Śodhana) and processing in Rasashastra. Zinc distillation techniques described in Rasārṇava and related texts. Concepts of acids and bases derived from plant, organic, and mineral sources. Transition from natural to mineral chemical systems. (7 hours)
- **Unit 6: Modern Scientific Interpretation and Relevance:** Interpretation of ancient chemical and metallurgical knowledge using modern scientific tools, frameworks, and data analysis, with connections to materials science, reaction mechanisms, and analytical techniques such as spectroscopy and surface analysis. Relevance to heritage materials, sustainable technologies, and contemporary materials research. (6 hours)

Laboratory/Practical/Tutorial Modules

NA

Text Books

1. Ray, P. C. (1902–1909). *A history of Hindu chemistry: From the earliest times to the middle of the sixteenth century A.D.* (Vols. 1-2).
2. Chattopadhyaya, D. (1986–1991). *History of science and technology in ancient India* (Vols. 1-3).

References

1. Ray, P. C. (1956). *History of Chemistry in Ancient and Medieval India*.
2. Biswas, A. K., & Biswas, S. (1995). *Minerals and Metals in Ancient India* (Vol. 2).
3. Agrawal, D. P. (2000). *Ancient metal technology and archaeology of South Asia*.
4. Skoog, D. A. *Principles of Instrumental Analysis*.
5. Brereton, R. G. *Chemometrics: Data Analysis for the Laboratory and Chemical Plant*.
6. *Rasaratna Samuccaya*.
7. *Rasārṇava*.
8. *Rasendra Chūḍāmaṇi*.